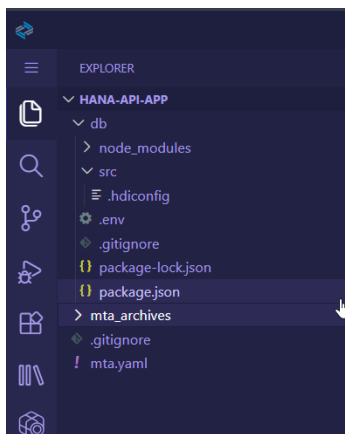


Application Development

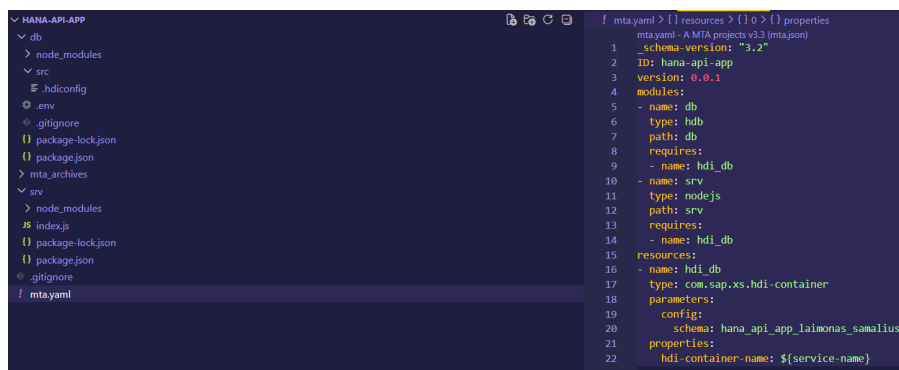
Link to solution, README.md, Reflection.md in GitHub:

<https://github.com/laimonuxlt/hana-api-app.git>

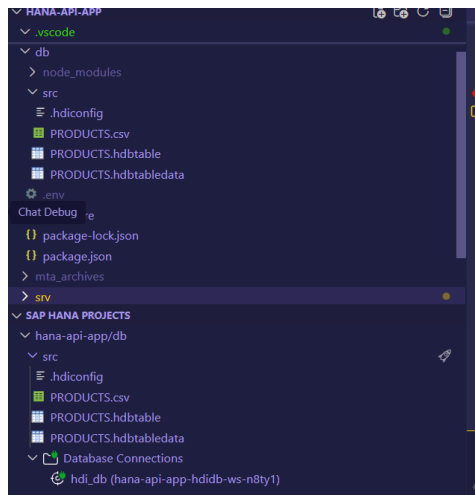
The screenshot of project structure:



The screenshot of database and Node.js modules:



The screenshot of deployed database objects:

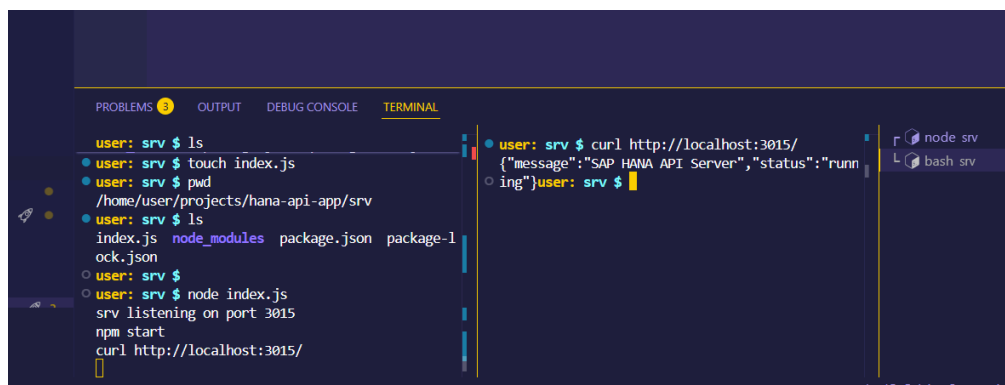


Sample data screenshot:

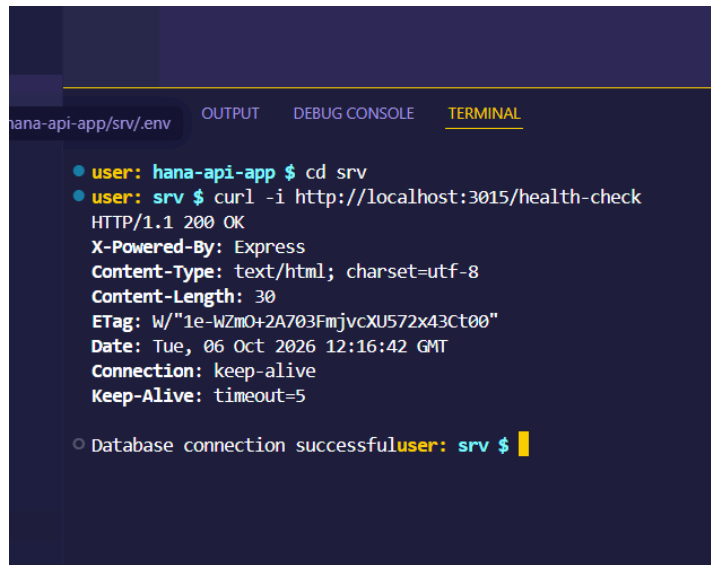
The screenshot shows the SAP HANA Studio interface with a SQL query executed. The query is: `1 - SELECT * FROM "HANA_API_APP_LAIMONAS_SAMALIUS_1"."PRODUCTS"`. The results are displayed in a table with 3 rows and 7 columns: ID, NAME, CATEGORY, PRICE, STOCK, and two unnamed columns.

	ID	NAME	CATEGORY	PRICE	STOCK		
1	1	Laptop	Electronics	999.00	25		
2	2	Desk Chair	Furniture	149.50	40		
3	3	Notebook	Stationery	3.20	500		

The screenshot of server running:



The screenshot of the successful connection:



```
hana-api-app/srv/.env OUTPUT DEBUG CONSOLE TERMINAL
● user: hana-api-app $ cd srv
● user: srv $ curl -i http://localhost:3015/health-check
HTTP/1.1 200 OK
X-Powered-By: Express
Content-Type: text/html; charset=utf-8
Content-Length: 30
ETag: W/"1e-WZm0+2A703FmjvcXU572x43Ct00"
Date: Tue, 06 Oct 2026 12:16:42 GMT
Connection: keep-alive
Keep-Alive: timeout=5

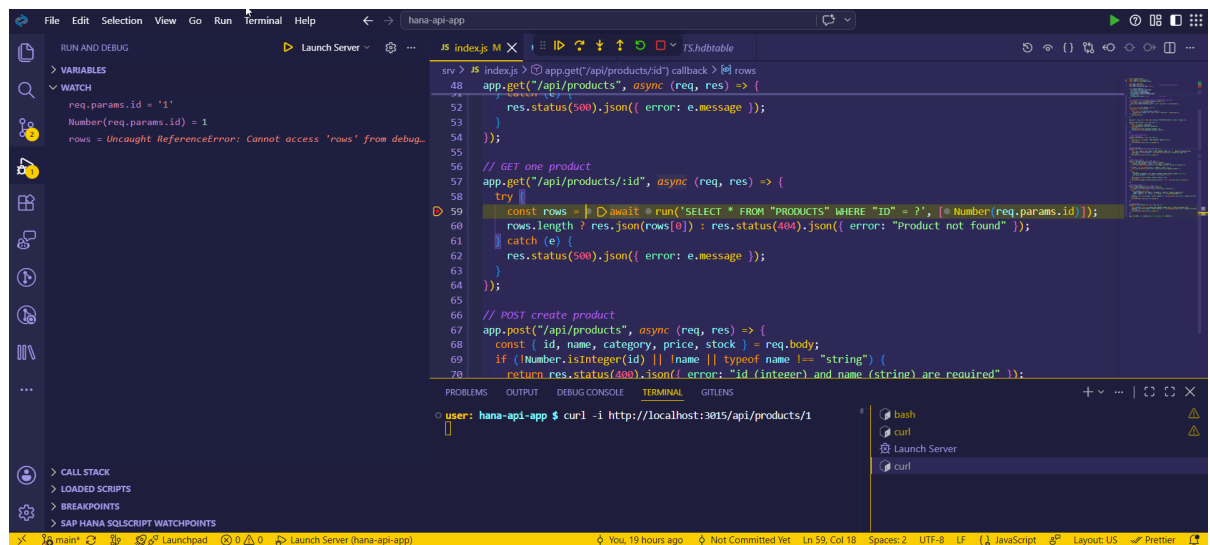
○ Database connection successfuluser: srv $
```

Screenshot of debugger in action:
Breakpoint set



```
54 });
55
56 // GET one product
57 app.get("/api/products/:id", async (req, res) => {
58   try {
59     const rows = await run('SELECT * FROM "PRODUCTS" WHERE "ID" = ?', [Number(req.params.id)]);
60     rows.length ? res.json(rows[0]) : res.status(404).json({ error: "Product not found" });
61   } catch (e) {
62     res.status(500).json({ error: e.message });
63   }
64 });
```

Execution paused



Variable inspection

The screenshot shows the VS Code interface with a JavaScript file named `index.js` open. The left sidebar displays the 'VARIABLES' and 'WATCH' panels. The 'WATCH' panel shows the following variables:

- `req.params.id = '1'`
- `Number(req.params.id) = 1`
- `rows = (1) [{}]`
- `0 = {ID: 1, NAME: 'Laptop', CATEGORY: 'Electronics', PRICE: '999.00', STOCK: 25}`
- `length = 1`
- `[[Prototype]] = Array(0)`
- `[[Prototype]] = Object`

The main editor shows the following code:

```
48 app.get("/api/products", async (req, res) => {
49   res.status(500).json({ error: e.message });
50 });
51
52 // GET one product
53 app.get("/api/products/:id", async (req, res) => {
54   try {
55     const rows = await e.run("SELECT * FROM 'PRODUCTS' WHERE 'ID' = ?", [Number(req.params.id)]);
56     rows.length ? res.json(rows[0]) : res.status(404).json({ error: "Product not found" });
57   } catch (e) {
58     res.status(500).json({ error: e.message });
59   }
60 });
61
62 // POST create product
63 app.post("/api/products", async (req, res) => {
64   const { id, name, category, price, stock } = req.body;
65   if (!Number.isInteger(id) || !name || typeof name !== "string") {
66     return res.status(400).json({ error: "id (integer) and name (string) are required" });
67   }
68   app.post("/api/products", async (req, res) => {
69     const { id, name, category, price, stock } = req.body;
70     if (!Number.isInteger(id) || !name || typeof name !== "string") {
71       return res.status(400).json({ error: "id (integer) and name (string) are required" });
72     }
73   });
74 });
```

API Testing

The screenshot shows the VS Code interface with a file named `request.http` open. The left sidebar shows the 'request.http' file. The main editor shows the following code:

```
1 Laimonas Samalius, 22 minutes ago | 1 author (Laimonas Samalius)
2 ### Health check
3 Send Request
4 GET http://localhost:3015/health-check
5
6 ### 1. GET all products
7 Send Request
8 GET http://localhost:3015/api/products
9
10 ### 2. POST - create a new product
11 Send Request
12 POST http://localhost:3015/api/products
```

The right sidebar shows the following output:

```
1 HTTP/1.1 200 OK
2 X-Powered-By: Express
3 Content-Type: text/html; charset=utf-8
4 Content-Length: 30
5 ETag: W/"1e-WZm0+2A703FmjvcXU572x43Ct00"
6 Date: Wed, 07 Oct 2026 14:13:48 GMT
7 Connection: close
8
9 Database connection successful
```

The screenshot shows the VS Code interface with a file named `request.http` open. The left sidebar shows the 'request.http' file. The main editor shows the following code:

```
1 Laimonas Samalius, 22 minutes ago | 1 author (Laimonas Samalius)
2 ### Health check
3 Send Request
4 GET http://localhost:3015/health-check
5
6 ### 1. GET all products
7 Send Request
8 GET http://localhost:3015/api/products
9
10 ### 2. POST - create a new product
11 Send Request
12 POST http://localhost:3015/api/products
13 Content-Type: application/json
14 {
15   "id": 100,
16   "name": "Monitor",
17   "category": "Electronics",
18   "price": 199.99,
19   "stock": 10
20 }
21
22 ### 3. GET all products (should include ID 100)
23 Send Request
24 GET http://localhost:3015/api/products
25
26 ### 4. GET single product by ID
27 Send Request
28 GET http://localhost:3015/api/products/100
29
30 ### 5. PUT - update the product
```

The right sidebar shows the following output:

```
1 HTTP/1.1 200 OK
2 X-Powered-By: Express
3 Content-Type: application/json; charset=utf-8
4 Content-Length: 237
5 ETag: W/"ed-VpPLK1IYK3y0jxHsJr4B98dlwg4"
6 Date: Wed, 07 Oct 2026 14:24:49 GMT
7 Connection: close
8
9 [
10   {
11     "ID": 1,
12     "NAME": "Laptop",
13     "CATEGORY": "Electronics",
14     "PRICE": "999.00",
15     "STOCK": 25
16   },
17   {
18     "ID": 2,
19     "NAME": "Desk Chair",
20     "CATEGORY": "Furniture",
21     "PRICE": "149.50",
22     "STOCK": 40
23   },
24   {
25     "ID": 3,
26     "NAME": "Notebook",
27     "CATEGORY": "Stationery",
```

```
index.js  request.http X  Welcome  Welcome  ...  Response(93ms) X
srv > request.http > POST /api/products
Laimonas Samalius, 22 minutes ago | 1 author (Laimonas Samalius)
1  ### Health check
  Send Request
2  GET http://localhost:3015/health-check
3
4  ### 1. GET all products
  Send Request
5  GET http://localhost:3015/api/products
6
7  ### 2. POST - create a new product
  Send Request
8  POST http://localhost:3015/api/products
9  Content-Type: application/json
10
11  {
12    "id": 100,
13    "name": "Monitor",
14    "category": "Electronics",
15    "price": 199.99,
16    "stock": 10
17  }
18

1  HTTP/1.1 201 Created
2  X-Powered-By: Express
3  Content-Type: application/json; charset=utf-8
4  Content-Length: 38
5  ETag: W/"26-AMnoe3n5Qs0uv9H2NLrpyDSTsrI"
6  Date: Wed, 07 Oct 2026 14:25:31 GMT
7  Connection: close
8
9  {
10    "message": "Product created",
11    "id": 100
12  }
```

```
request.http > GET /api/products
Laimonas Samalius, 22 minutes ago | 1 author (Laimonas Samalius)
### Health check
Send Request
GET http://localhost:3015/health-check

### 1. GET all products
Send Request
GET http://localhost:3015/api/products

### 2. POST - create a new product
Send Request
POST http://localhost:3015/api/products
Content-Type: application/json

{
  "id": 100,
  "name": "Monitor",
  "category": "Electronics",
  "price": 199.99,
  "stock": 10
}

### 3. GET all products (should include ID 100)
Send Request
GET http://localhost:3015/api/products

### 4. GET single product by ID
Send Request
GET http://localhost:3015/api/products/100

### 5. PUT - update the product

9  {
10  {
11    "ID": 1,
12    "NAME": "Laptop",
13    "CATEGORY": "Electronics",
14    "PRICE": "999.00",
15    "STOCK": 25
16  },
17  {
18    "ID": 2,
19    "NAME": "Desk Chair",
20    "CATEGORY": "Furniture",
21    "PRICE": "149.50",
22    "STOCK": 40
23  },
24  {
25    "ID": 3,
26    "NAME": "Notebook",
27    "CATEGORY": "Stationery",
28    "PRICE": "3.20",
29    "STOCK": 500
30  },
31  {
32    "ID": 100,
33    "NAME": "Monitor",
34    "CATEGORY": "Electronics",
35    "PRICE": "199.99",
36    "STOCK": 10
37  }
```

```
JS index.js  request.http X Welcome Welcome ... Response(98ms) X
srv > request.http > GET /api/products/100
2 GET http://localhost:3015/health-check
3
4 ### 1. GET all products
5 Send Request
6 GET http://localhost:3015/api/products
7
8 ### 2. POST - create a new product
9 Send Request
10 POST http://localhost:3015/api/products
11 Content-Type: application/json
12 {
13   "id": 100,
14   "name": "Monitor",
15   "category": "Electronics",
16   "price": 199.99,
17   "stock": 10
18 }
19
20 ### 3. GET all products (should include ID 100)
21 Send Request
22 GET http://localhost:3015/api/products
23
24 ### 4. GET single product by ID
25 Send Request
26 GET http://localhost:3015/api/products/100
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